Prem Seetharaman

Contact and Info
954-558-3546
56seeth@gmail.com
pseeth.github.io
github.com/pseeth

Research interests

Lowering the barrier to entry for creative work in video and audio by leveraging and developing state-of-the-art generative modeling techniques. My research interests are primarily in computer audition. My recent work spans media understanding and generation, especially for audio (sound effects, music, and speech), as well as audiovisual generation. I am also interested in audio representation learning, audio compression, and audio-to-audio mapping (e.g. speech/music separation and enhancement).

Education

Northwestern University, Evanston, IL
09/2019 - PhD, Computer Science

Northwestern University, Evanston, IL
BS, Computer Science, Music Composition - 2013

Professional employment

Adobe Research
Sr. Research Scientist: 2023 - Present

- Generate Soundscape in Adobe Premiere (beta) - picture-synced soundscapes for video, based on our MultiFoley research.
  https://community.adobe.com/announcements-732/now-in-beta-generate-soundscape-in-premiere-1639274
- Generate Sound Effects in Adobe Firefly (beta) - sound effects from text and vocal imitation, based on our Sketch2Sound research.
  https://helpx.adobe.com/firefly/web/work-with-audio-and-video/work-with-audio/voice-to-sound-effects.html
- Contributed to Adobe MAX Sneaks: Project Super Sonic (2024) and Project Sound Stager (2025).

Descript, Inc.
Research Scientist: 2020 - 2023

- Led Studio Sound - state-of-the-art speech enhancement.
  https://www.descript.com/studio-sound
- Core contributor to Overdub - voice cloning for speech editing.
  https://www.descript.com/overdub
- Developed the Descript Audio Codec - high-fidelity audio compression, which powers internal audio language models for editing speech and generating music.
  https://github.com/descriptinc/descript-audio-codec

Northwestern University, Evanston, IL
Postdoctoral Scholar: 2019 - 2020

Northwestern University, Evanston, IL
Doctoral Student in Interactive Audio Lab: 2013 - 2019

Mitsubishi Electric Research Labs, Cambridge, MA
Research Intern: 2018

Adobe Research, San Francisco, CA
Research Intern: 2017-2018

Gracenote, Emeryville, CA
Research Intern: 2016

Publications

Thesis

[1] **Seetharaman, Prem**. "Bootstrapping the Learning Process for Computer Audition". PhD thesis. Northwestern University, 2019.

Patents

[1] Prem Seetharaman, Oriol Nieto, and Justin Salamon. *Generative Audio Extension*. US Patent App. 18/916,334. 2026.

[2] Justin Salamon, Prem Seetharaman, Oriol Nieto, Hugo Flores García, et al. *Generative Sound Effects for Video Editing*. US Patent App. 18/916,465. 2026.

[3] Prem Seetharaman and Kundan Kumar. *Training Machine Learning Frameworks to Generate Studio-Quality Recordings Through Manipulation of Noisy Audio Signals*. US Patent 12,431,154. 2025.

[4] Zafar Rafii and Prem Seetharaman. *Audio Identification Based on Data Structure*. US Patent 12,105,754. 2024.

[5] Prem Seetharaman and Kundan Kumar. *Approaches to Generating Studio-Quality Recordings Through Manipulation of Noisy Audio*. US Patent App. 18/154,718. July 2023.

[6] Prem Seetharaman, Gautham J Mysore, and Bryan A Pardo. *Sound Quality Prediction and Interface to Facilitate High-Quality Voice Recordings*. US Patent App. 16/296,122. Sept. 2020.

[7] Markus K Cremer, Zafar Rafii, Robert Coover, and Prem Seetharaman. *Automated Cover Song Identification*. US Patent App. 15/698,557. July 2018.

Journal Articles

[1] Vivian Tang, Prem Seetharaman, Kevin Chao, Bryan A Pardo, and Suzan Van Der Lee. "Automating the detection of dynamically triggered earthquakes via a deep metric learning algorithm". In: *Seismological Research Letters* 91.2A (2020), pp. 901–912.

[2] Bryan Pardo, Mark Cartwright, Prem Seetharaman, and Bongjun Kim. "Learning to build natural audio production interfaces". In: *Arts*. Vol. 8. 3. MDPI. 2019, p. 110.

[3] Eric J Humphrey, Sravana Reddy, Prem Seetharaman, Aparna Kumar, Rachel M Bittner, Andrew Demetriou, Sankalp Gulati, Andreas Jansson, Tristan Jehan, Bernhard Lehner, et al. "An introduction to signal processing for singing-voice analysis: High notes in the effort to automate the understanding of vocals in music". In: *IEEE Signal Processing Magazine* 36.1 (2018), pp. 82–94.

[4] Prem Seetharaman and Bryan Pardo. "Audealize: Crowdsourced Audio Production Tools". In: *Journal of the Audio Engineering Society* 64.9 (2016), pp. 683–695.

Papers

[1] William Chen, Prem Seetharaman, Ke Chen, Oriol Nieto, Kevin Duarte, Siddharth Srinivasan Iyer, Mamshad Nayeem Rizve, Zhiwen Cao, Shinji Watanabe, Yuanjun Xiong, Jianming Zhang, Zeyu Jin, and Justin Salamon. "CrossEdit: Cross-Modal Training Enables Rich Audio-Visual Editing". In: *arXiv preprint arXiv:2610.10264* (2026). URL: https://arxiv.org/abs/2610.10264.

[2] Sonal Kumar, Prem Seetharaman, Ke Chen, Oriol Nieto, Jiaqi Su, Zhepei Wang, Rithesh Kumar, Dinesh Manocha, Nicholas J. Bryan, Zeyu Jin, and Justin Salamon. "TAC: Timestamped Audio Captioning". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.

[3] Annie Chu, Hugo Flores García, Johannes Imort, Oriol Nieto, Bryan Pardo, Jordan Rudess, Prem Seetharaman, and Justin Salamon. "Smorph: Playable Sound Morphing with Diffusion Models". In: *Proceedings of the 27th International Society for Music Information Retrieval Conference*. 2026.

[4] Ashish Seth, Sonal Kumar, Ramaneswaran Selvakumar, Nishit Anand, Utkarsh Tyagi, Prem Seetharaman, Ramani Duraiswami, and Dinesh Manocha. "Audio Hallucination Attacks: Probing the Reliability of Large Audio Language Models". In: *Proceedings of Interspeech*. 2026.

[5] William Chen, Prem Seetharaman, Rithesh Kumar, Oriol Nieto, Shinji Watanabe, Justin Salamon, and Zeyu Jin. "AudioChat: Unified Audio Storytelling, Editing, and Understanding with Transfusion Forcing". In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026.

[6] Prem Seetharaman and Rithesh Kumar. "Taming Audio VAEs via Target-KL Regularization". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026.

[7] Prem Seetharaman, Oriol Nieto, and Justin Salamon. "Generative Audio Extension and Morphing". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026.

[8] Annie Chu, Hugo Flores García, Oriol Nieto, Justin Salamon, Bryan Pardo, and Prem Seetharaman. "Mix2Morph: Learning Sound Morphing from Noisy Mixes". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026.

[9] Sripathi Sridhar, Prem Seetharaman, Oriol Nieto, Mark Cartwright, and Justin Salamon. "AudioCards: Structured Metadata Improves Audio Language Models for Sound Design". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026.

[10] Yutong Wen, Ke Chen, Prem Seetharaman, Oriol Nieto, Jiaqi Su, Rithesh Kumar, Minje Kim, Paris Smaragdis, Zeyu Jin, and Justin Salamon. "PromptSep: Generative Audio Separation via Multimodal Prompting". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026.

[11] Jialin Huang, Prem Seetharaman, Tim Langlois, Li-Yi Wei, Rubaiat Habib Kazi, and Yotam Gingold. "MoSound: An Interactive Tool for Generative Sound Design in Motion Graphics". In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 2026. **Best Paper Honorable Mention**.

[12] Suhyeon Yoo, Adolfo Hernandez Santisteban, Prem Seetharaman, Justin Salamon, Oriol Nieto, and Anh Truong. "SoundStager: Interactive Design of Story-Driven GenAI Soundscapes for Video". In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 2026.

[13] Patrick O'Reilly, Julia Barnett, Hugo Flores García, Annie Chu, Nathan Pruyne, Prem Seetharaman, and Bryan Pardo. "The Rhythm In Anything: Audio-Prompted Drums Generation with Masked Language Modeling". In: *Proceedings of the 26th International Society for Music Information Retrieval Conference*. 2025.

[14] Sonal Kumar, Prem Seetharaman, Justin Salamon, Dinesh Manocha, and Oriol Nieto. "SILA: Signal-to-Language Augmentation for Enhanced Control in Text-to-Audio Generation". In: *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*. 2025.

[15] Yusong Wu, Christos Tsirigotis, Ke Chen, Cheng-Zhi Anna Huang, Aaron Courville, Oriol Nieto, Prem Seetharaman, and Justin Salamon. "FLAM: Frame-Wise Language-Audio Modeling". In: *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. 2025.

[16] Ziyang Chen, Prem Seetharaman, Bryan Russell, Oriol Nieto, David Bourgin, Andrew Owens, and Justin Salamon. "Video-Guided Foley Sound Generation with Multimodal Controls". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025.

[17] Hugo Flores García, Oriol Nieto, Justin Salamon, Bryan Pardo, and Prem Seetharaman. "Sketch2Sound: Controllable Audio Generation via Time-Varying Signals and Sonic Imitations". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025.

[18] Patrick O'Reilly, Prem Seetharaman, Jiaqi Su, Zeyu Jin, and Bryan Pardo. "Code Drift: Towards Idempotent Neural Audio Codecs". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025.

[19] Hugo Flores Garcia, Prem Seetharaman, Rithesh Kumar, and Bryan Pardo. "VampNet: Music Generation via Masked Acoustic Token Modeling". In: *Proceedings of the 24th International Society for Music Information Retrieval Conference*. 2023.

[20] Rithesh Kumar, Prem Seetharaman, Alejandro Luebs, Ishaan Kumar, and Kundan Kumar. "High-Fidelity Audio Compression with Improved RVQGAN". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.

[21] Noah Schaffer, Boaz Cogan, Ethan Manilow, Max Morrison, Prem Seetharaman, and Bryan Pardo. "Music Separation Enhancement with Generative Modeling". In: (2022).

[22] Ho Hsiang Wu, Magdalena Fuentes, Prem Seetharaman, and Juan Pablo Bello. "How to Listen? Rethinking Visual Sound Localization". In: *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. Vol. 2022. 2022, pp. 876–880.

[23] Ho-Hsiang Wu, Prem Seetharaman, Kundan Kumar, and Juan Pablo Bello. "Wav2clip: Learning robust audio representations from clip". In: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2022, pp. 4563–4567.

[24] Ethan Manilow, Patrick O'Reilly, Prem Seetharaman, and Bryan Pardo. "Source separation by steering pretrained music models". In: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2022, pp. 126–130.

[25] Max Morrison, Rithesh Kumar, Kundan Kumar, Prem Seetharaman, Aaron Courville, and Yoshua Bengio. "Chunked autoregressive GAN for conditional waveform synthesis". In: *International Conference on Learning Representations*. 2021.

[26] Nicolas Turpault, Romain Serizel, Scott Wisdom, Hakan Erdogan, John R Hershey, Eduardo Fonseca, Prem Seetharaman, and Justin Salamon. "Sound event detection and separation: a benchmark on desed synthetic soundscapes". In: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2021, pp. 840–844.

[27] Andreas Bugler, Bryan Pardo, and Prem Seetharaman. "A study of transfer learning in music source separation". In: *arXiv preprint arXiv:2010.12650* (2020).

[28] Prem Seetharaman, Gordon Wichern, Bryan Pardo, and Jonathan Le Roux. "AutoClip: Adaptive Gradient Clipping for Source Separation Networks". In: *2020 IEEE 30th International Workshop on Machine Learning for Signal Processing (MLSP)*. IEEE. 2020, pp. 1–6.

[29] Omkar Ranadive, Grant Gasser, David Terpay, and Prem Seetharaman. "Otoworld: Towards learning to separate by learning to move". In: *ICML 2020 Workshop on Self-supervision in Audio and Speech*. 2020.

[30] Scott Wisdom, Hakan Erdogan, Daniel PW Ellis, Romain Serizel, Nicolas Turpault, Eduardo Fonseca, Justin Salamon, Prem Seetharaman, and John R Hershey. "What's all the fuss about free universal sound separation data?" In: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2021, pp. 186–190.

[31] Nicolas Turpault, Scott Wisdom, Hakan Erdogan, John R Hershey, Romain Serizel, Eduardo Fonseca, Prem Seetharaman, and Justin Salamon. "Improving Sound Event Detection In Domestic Environments Using Sound Separation". In: *DCASE Workshop-Detection and Classification of Acoustic Scenes and Events 2020*. 2020.

[32] Alisa Liu, Alexander Fang, Gaëtan Hadjeres, Prem Seetharaman, and Bryan Pardo. "Incorporating Music Knowledge in Continual Dataset Augmentation for Music Generation". In: *ML4MD Workshop at ICML 2020*. 2020.

[33] Alexander Fang, Alisa Liu, Prem Seetharaman, and Bryan Pardo. "Bach or mock? a grading function for chorales in the style of JS Bach". In: *ML4MD Workshop at ICML 2020*. 2020.

[34] Prem Seetharaman, Gordon Wichern, Jonathan Le Roux, and Bryan Pardo. "Bootstrapping unsupervised deep music separation from primitive auditory grouping principles". In: *ICML 2020 Workshop on Self-supervision in Audio and Speech*. 2020.

[35] Ethan Manilow, Prem Seetharaman, and Bryan Pardo. "Simultaneous separation and transcription of mixtures with multiple polyphonic and percussive instruments". In: *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2020, pp. 771–775.

[36] Fatemeh Pishdadian, Bongjun Kim, Prem Seetharaman, and Bryan Pardo. "Classifying Non-Speech Vocals: Deep vs Signal Processing Representations". In: 2019.

[37] Ethan Manilow, Gordon Wichern, Prem Seetharaman, and Jonathan Le Roux. "Cutting Music Source Separation Some Slakh: A Dataset to Study the Impact of Training Data Quality and Quantity". In: *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*. 2019.

[38] Prem Seetharaman. "Bootstrapping the Learning Process for Computer Audition". PhD thesis. Northwestern University, 2019.

[39] Alisa Liu, Prem Seetharaman, and Bryan Pardo. "Model Selection for Deep Audio Source Separation via Clustering Analysis". In: 2020.

[40] Prem Seetharaman, Gautham Mysore, Bryan Pardo, Paris Smaragdis, and Celso Gomes. "VoiceAssist: Guiding Users to High-Quality Voice Recordings". In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM. 2019.

[41] Prem Seetharaman, Gordon Wichern, Shrikant Venkataramani, and Jonathan Le Roux. "Class-Conditional Embeddings for Music Source Separation". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019*. 2018.

[42] Prem Seetharaman, Gordon Wichern, Jonathan Le Roux, and Bryan Pardo. "Bootstrapping Single-Channel Source Separation via Unsupervised Spatial Clustering on Stereo Mixtures". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019*. 2018.
[43] Ethan Manilow, Prem Seetharaman, Fatemah Pishdadian, and Bryan Pardo. *NUSSL: the northwestern university source separation library*. 2018.
[44] Julia Wilkins, Prem Seetharaman, Alison Wahl, and Bryan Pardo. "VocalSet: A Singing Voice Dataset". In: *Proceedings of the 19th International Society for Music Information Retrieval Conference*. 2018.
[45] Prem Seetharaman, Gautham J Mysore, Paris Smaragdis, and Bryan Pardo. "Blind Estimation of the Speech Transmission Index for Speech Quality Prediction". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018*. 2018.
[46] Prem Seetharaman, Fatemeh Pishdadian, and Bryan Pardo. "Music/Voice Separation Using the 2D Fourier Transform". In: *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*. 2017.
[47] Ethan Manilow, Prem Seetharaman, Fatemeh Pishdadian, and Bryan Pardo. "Predicting Algorithm Efficacy for Adaptive Multi-Cue Source Separation". In: *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*. 2017.
[48] Prem Seetharaman and Zafar Rafii. "Cover Song Identification with 2D Fourier Transform Sequences". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2017*. 2017.
[49] Taylor Zheng, Prem Seetharaman, and Bryan Pardo. "SocialFX: Studying a Crowdsourced Folksonomy of Audio Effects Terms". In: *Proceedings of the 2016 ACM on Multimedia Conference*. ACM. 2016, pp. 182–186.
[50] Prem Seetharaman and Bryan Pardo. "Simultaneous Separation and Segmentation in Layered Music". In: *Proceedings of the 17th International Society for Music Information Retrieval Conference*. 2016, pp. 495–502.
[51] Prem Seetharaman and Bryan Pardo. "Crowdsourcing a Reverberation Descriptor Map". In: *Proceedings of the 22nd ACM international conference on Multimedia*. ACM. 2014, pp. 587–596.
[52] Prem Seetharaman and Stephen P Tarzia. "The Hand Clap as an Impulse Source for Measuring Room Acoustics". In: *Audio Engineering Society Convention 132*. Audio Engineering Society. 2012.

Extended Abstracts

[1] Vivian Tang, Prem Seetharaman, Kevin Chao, Bryan A Pardo, and Suzan van der Lee. "Siamese networks for triggered earthquakes detection". In: *AGU Fall Meeting Abstracts*. Vol. 2018. 2018, S11E–0428.
[2] Kevin Chao, Prem Seetharaman, Vivian Tang, Bryan A Pardo, and Suzan Van der Lee. "Automatic classification of triggered tectonic tremor with deep learning". In: *AGU Fall Meeting Abstracts*. Vol. 2018. 2018, T33E–0463.
[3] Prem Seetharaman and Bryan Pardo. "Reverbalize: A Crowdsourced Reverberation Controller". In: *Proceedings of the 22nd ACM international conference on Multimedia*. ACM. 2014, pp. 739–740.
[4] Michael Donovan, Prem Seetharaman, and Bryan Pardo. "A Web Audio Node for the Fast Creation of Natural Language Interfaces for Audio Production". In: (2017).

References

Bryan Pardo
Full Professor
Department of Computer Science, Northwestern University
Evanston, IL
847.491.7184
pardo@northwestern.edu

Ken Forbus
Walter P. Murphy Professor of Computer Science
Department of Computer Science, Northwestern University
847.491.7699
forbus@northwestern.edu

Paris Smaragdis

Associate Professor
Department of Computer Science, University of Illinois at Urbana Champaign
217.265.6893
paris@illinois.edu

Sara Sood
Chookaszian Family Teaching Professor, Northwestern University
847.491.5708
sara@northwestern.edu

Gautham Mysore
Principal Scientist, Adobe Research
Adjunct Professor, Stanford University
650.353.1142
gmysore@adobe.com

Jonathan Le Roux
Senior Principal Research Scientist, Mitsubishi Electric Research Laboratories (MERL)
617.621.7547
leroux@merl.com